INVERTIS UNIVERSITY, BAREILLY (U.P.)

Faculty of Computer Applications • Department of BCA

🎓 **FINAL YEAR CAPSTONE PROJECT DOCUMENTATION**

ITC Hotels: Revenue & Business Performance Analysis

Complete Technical Architecture & File-by-File Documentation Manual

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Academic Session: [2025–26] • Repository: https://github.com/Avis1229/ITC_HOTEL_S_MODULE

1. Executive Summary & Key Business Metrics

This project delivers a full-stack, enterprise-grade business intelligence and database system modeled around **ITC Hotels**, India's premier luxury chain renowned for its *Responsible Luxury* ethos. Analyzing **119,390 historical reservations** across luxury city hotels (*ITC Maurya, ITC Grand Chola*) and resorts (*ITC Grand Bharat, ITC Mughal*), the project bridges transactional relational database operations with advanced analytics across 4 core business domains.

GROSS REVENUE	GROSS OPERATING PROFIT	AVERAGE DAILY RATE	F&B + BANQUETS
₹ 298.42 Cr	₹ 143.68 Cr	₹ 8,941	₹ 104.84 Cr
• 119k Total Bookings	48.15% Margin (USALI)	Per Room Night Realized	Bukhara / Dum Pukht / MICE

The system completely replaces legacy, manual registers with a **3NF Normalized Relational Database** (18 tables & views, 87,396 cleaned transactions), a transparent **Python data cleaning & exploratory data analysis (EDA) pipeline** with high-resolution Matplotlib visualizations, a **Master Microsoft Excel live dashboard** with 5 tabs and square KPI cards, and an enterprise **Power BI Direct SQL Integration Model** with industry-standard DAX measures.

2. Master Project File Directory & Role Index

Below is the complete inventory of all 42 files across the project workspace, categorized by operational role:

DIRECTORY / FILE PATH	TYPE	PRIMARY FUNCTION & PURPOSE
`README.md`	DOCUMENTATION	Master GitHub landing page containing project abstract, team details, architecture diagram, and branch index.
`PROJECT_SUBMISSION_CHECKLIST.md`	DOCUMENTATION	Academic evaluation rubric, viva defense questions, file inventory, and submission verification steps.
`run_dashboard.sh`	AUTOMATION	Interactive terminal launcher menu (Jupyter EDA, Plotly dashboard, Excel, SQLite shell, table verification).
`push_all_to_github.sh`	AUTOMATION	Bash automation script for configuring Git remotes and pushing all 5 isolated module branches to GitHub.
`.gitignore`	CONFIG	Protects repository integrity by ignoring `.venv/`, Python bytecode, OS lock files, and raw dumps.

DIRECTORY / FILE PATH	TYPE	PRIMARY FUNCTION & PURPOSE
`archive/hotel_bookings.csv`	RAW DATA	Original Kaggle Hotel Booking Demand dataset (119,390 rows × 32 columns) used as the ingestion baseline.
`ITC-Room-Analytics/itc_room_revenue_occupancy.csv`	CURATED DATA	Module 1 dataset (119k rows × 21 cols): room category mapping, ADR in INR, length of stay, lead times.
`ITC-Room-Analytics/module1_room_revenue_occupancy.ipynb`	JUPYTER	Executed Module 1 analysis notebook: direct SQL database connection, RevPAR curves, category yield, Matplotlib charts.
`ITC-Room-Analytics/README.md`	DOCUMENTATION	Module 1 specification, academic headers, KPI summaries, and reproduction instructions.
`ITC-Room-Analytics/requirements.txt`	CONFIG	Python dependencies for running Module 1 (pandas, numpy, matplotlib, seaborn).
`ITC-FnB-Analytics/itc_fnb_banquet_performance.csv`	CURATED DATA	Module 2 dataset (119k rows × 17 cols): restaurant outlets (Bukhara, Dum Pukht), dining covers, banquet events.
`ITC-FnB-Analytics/module2_fnb_banquet_performance.ipynb`	JUPYTER	Executed Module 2 analysis notebook: SQL query loading, covers revenue, dining plan margins, banquet event yields.
`ITC-FnB-Analytics/README.md`	DOCUMENTATION	Module 2 specification, culinary outlet benchmarks, and execution steps.
`ITC-FnB-Analytics/requirements.txt`	CONFIG	Python dependencies for running Module 2.
`ITC-Guest-Demographics/itc_guest_demographics_channels.csv`	CURATED DATA	Module 3 dataset (119k rows × 22 cols): guest origin country, personas, Club ITC loyalty tiers, distribution channels.
`ITC-Guest-Demographics/module3_guest_demographics_channels.ipynb`	JUPYTER	Executed Module 3 analysis notebook: SQL queries, channel risk analysis, repeat guest loyalty retention.
`ITC-Guest-Demographics/README.md`	DOCUMENTATION	Module 3 specification, customer segmentation overview, and run guide.

DIRECTORY / FILE PATH	TYPE	PRIMARY FUNCTION & PURPOSE
`ITC-Guest-Demographics/requirements.txt`	CONFIG	Python dependencies for running Module 3.
`ITC-Profitability-Costs/itc_operating_costs_profitability.csv`	CURATED DATA	Module 4 dataset (119k rows × 21 cols): USALI hotel P&L accounting, departmental COGS, staff payroll, utilities, GOP.
`ITC-Profitability-Costs/module4_operating_costs_profitability.ipynb`	JUPYTER	Executed Module 4 notebook: SQL aggregation, USALI cost center breakdown, GOP margin benchmarking.
`ITC-Profitability-Costs/README.md`	DOCUMENTATION	Module 4 specification, USALI standard definitions, and KPI calculations.
`ITC-Profitability-Costs/requirements.txt`	CONFIG	Python dependencies for running Module 4.
`ITC-Final-Dashboards/itc_hotel_analytics.db`	DATABASE	3NF Normalized SQLite enterprise database (81 MB, 18 tables & views, 87,396 cleaned transactions).
`ITC-Final-Dashboards/itc_hotels_mysql_dump.sql`	DATABASE	Complete MySQL 8.0 DDL & Data Dump script ready for instant import into XAMPP, phpMyAdmin, or cloud MySQL.
`ITC-Final-Dashboards/itc_master_hospitality_data.csv`	MASTER DATA	Consolidated 74-column master dataset combining room, F&B, banquet, demographics, and USALI financial metrics.
`ITC-Final-Dashboards/data_cleaning_eda.ipynb`	JUPYTER	Comprehensive data cleaning pipeline & Matplotlib EDA notebook with live SQL query integration.
`ITC-Final-Dashboards/interactive_jupyter_dashboard.ipynb`	JUPYTER	Interactive executive dashboard in Plotly Dark Theme featuring 5 tabs, 4 KPI cards, and waterfall charts.
`ITC-Final-Dashboards/ITC_Executive_Live_Dashboard.xlsx`	EXCEL	Master Microsoft Excel live dashboard featuring 5 bottom tabs, top square KPI cards, and native embedded charts.
`ITC-Final-Dashboards/itc_luxury_theme.json`	THEME	Official Power BI Dark Theme JSON preset styled with ITC luxury colors (Slate, Gold, Emerald, Ice Blue).

DIRECTORY / FILE PATH	TYPE	PRIMARY FUNCTION & PURPOSE
`ITC-Final-Dashboards/POWER_BI_SQL_INTEGRATION_GUIDE.md`	DOCUMENTATION	Comprehensive guide to connecting Power BI to the SQL database, Star Schema modeling, and official DAX measures.
`ITC-Final-Dashboards/POWER_BI_MODELING_GUIDE.md`	DOCUMENTATION	Step-by-step layout guide for creating the 5 visual pages matching the university report screenshots.
`ITC-Final-Dashboards/PROJECT_VIVA_PREPARATION.md`	DOCUMENTATION	High-yield viva revision guide with expected technical and domain questions, answers, and formulas.
`ITC-Final-Dashboards/README.md`	DOCUMENTATION	Executive suite documentation overview, repository layout, and usage instructions.
`ITC-Final-Dashboards/requirements.txt`	CONFIG	Dependencies for running the master dashboards (plotly, openpyxl, ipywidgets, matplotlib).
`scripts/process_data.py`	SCRIPT	ETL pipeline converting raw Kaggle data into INR luxury context and generating the 4 module CSVs.
`scripts/create_database.py`	SCRIPT	Cleans raw data (nulls, duplicates, outliers), constructs the 12 core 3NF tables, and populates SQLite.
`scripts/build_pdf_schema_and_mysql.py`	SCRIPT	Adds PDF-matched tables (properties, rooms, bookings, inventory) and exports `itc_hotels_mysql_dump.sql`.
`scripts/generate_eda_notebook.py`	SCRIPT	Generates the standalone `data_cleaning_eda.ipynb` notebook showing data cleaning and Matplotlib plots.
`scripts/generate_notebooks.py`	SCRIPT	Builds all 4 module notebooks with SQL queries, Invertis academic headers, and Seaborn/Matplotlib plots.
`scripts/create_excel_dashboard.py`	SCRIPT	Automates Openpyxl creation of the 5-sheet live Microsoft Excel executive dashboard with native charts.
`scripts/generate_jupyter_dashboard.py`	SCRIPT	Generates the Plotly dark-themed interactive executive dashboard notebook with tabbed widgets.

DIRECTORY / FILE PATH	TYPE	PRIMARY FUNCTION & PURPOSE
`scripts/setup_git_and_readmes.py`	SCRIPT	Configures git author information, initializes module repositories, and sets up branch structures.
`scripts/update_readmes.py`	SCRIPT	Synchronizes project team names, Invertis University metadata, and guide details across all README files.

3. File-by-File Technical Deep Dive

1. README.md (Root Project Landing Page)

DOCUMENTATION

The master entrypoint for the GitHub repository. It presents a comprehensive overview of the capstone project, formally crediting the Invertis University student team and guide Ms. Megha Saxena. It features executive KPI highlights, an interactive branch navigation table, an ASCII directory tree, database architectural highlights, and local quick-start instructions.

FILE SIZE:	LANGUAGE:	AUDIENCE:
6.2 KB	Markdown (GFM)	Evaluators, Recruiters, Faculty

2. PROJECT_SUBMISSION_CHECKLIST.md

ACADEMIC EVALUATION

Serves as an exhaustive self-audit checklist and evaluation guide for academic presentation. It outlines the 4 analytical modules, summarizes core metrics, cross-references files with the university project report, and details high-priority viva questions (e.g., USALI cost modeling, RevPAR formulas, star schema benefits, and data hygiene rules).

FILE SIZE:	KEY FOCUS:	SECTIONS:
9.8 KB	Viva Readiness & Verification	7 Structured Evaluation Units

3. run_dashboard.sh (Terminal Hub)

BASH SHELL SCRIPT

A menu-driven interactive terminal hub written in Bash. Designed specifically for the Ubuntu development environment, it allows the user or examiner to launch any component with a single keystroke: (1) Launch Data Cleaning & Matplotlib EDA in Jupyter; (2) Launch Plotly Executive Dashboard; (3) Open Excel Live Dashboard via LibreOffice/Excel; (4) Open Interactive SQLite CLI Shell on `itc_hotel_analytics.db`; (5) Display Real-time Table Record Counts across all 18 database entities.

PERMISSIONS:	INTERPRETER:	DEPENDENCIES:
Executable (755)	`/usr/bin/env bash`	python3, jupyter, sqlite3

4. push_all_to_github.sh

GIT AUTOMATION

Automates multi-module branch pushing to GitHub. When provided with the target GitHub username (e.g., `./push_all_to_github.sh Avis1229`), it systematically iterates through all module directories, attaches the appropriate remote origin URL, sets the default branch to `main`, and pushes isolated commits.

PERMISSIONS:	TARGET REPO:	ERROR HANDLING:
Executable (755)	`ITC_HOTEL_S_MODULE.git`	`set -e` Strict Mode

5. archive/hotel_bookings.csv (Raw Master Data)

CSV DATASET

The raw, unmodified benchmark Kaggle dataset containing 119,390 hotel booking records spanning 2015 to 2017. It comprises 32 original attributes including arrival dates, lead times, customer segments, room types, ADR in Euros, and cancellation flags. This file serves as the raw immutable ground truth from which all downstream pipelines derive.

RECORD COUNT:	FILE SIZE:	ORIGIN:
119,390 rows × 32 cols	16.8 MB	Kaggle (Jesse Mostipak)

3.1 Module 1: Room Revenue & Occupancy Analytics (`ITC-Room-Analytics`)

6. ITC-Room-Analytics/itc_room_revenue_occupancy.csv

MODULE DATASET

Cleaned transactional dataset specifically tailored for room revenue operations. Raw ADR in Euros is transformed to INR (conversion factor × 85), room categories are mapped to authentic ITC suites (*ITC One Luxury Room, Towers Club, Executive Suite, Presidential Suite*), and total stay nights are computed.

DIMENSIONS:	KEY COLUMNS:	FILE SIZE:
119,390 rows × 21 cols	adr_inr, room_revenue_inr, total_nights	22.1 MB

7. ITC-Room-Analytics/module1_room_revenue_occupancy.ipynb

JUPYTER NOTEBOOK

Executed diagnostic analytics notebook. It first executes a direct SQL query against `itc_hotel_analytics.db` to retrieve reservations and invoices, performs dataset hygiene checks, and generates publication-quality Seaborn and Matplotlib figures: (1) Room Revenue by Property Type (City Hotel vs Luxury Resort); (2) ADR Distribution Boxplot; (3) Revenue by Assigned Room Category; (4) Monthly Demand & Cancellation Seasonality Dual-Axis Curves.

EXECUTION STATE:	VISUALS:	OUTPUT SIZE:
Fully Pre-Executed	Bar, Boxplot, Dual-Axis Line	279 KB

3.2 Module 2: Food, Beverage & Banquet Performance (`ITC-FnB-Analytics`)

8. ITC-FnB-Analytics/itc_fnb_banquet_performance.csv

MODULE DATASET

Hospitality dining and banquet performance dataset. It models realistic covers and spending across ITC's legendary signature restaurants (*Bukhara, Dum Pukht, Pavilion, Pan Asian, West View, Peshawri*) and MICE venues (*The Grand Ballroom, Peacock Hall, Mughal Hall*). It separates food sales, beverage sales, and banquet attendee yields.

DIMENSIONS:	KEY OUTLETS:	TOTAL PORTFOLIO YIELD:
119,390 rows × 17 cols	Bukhara, Dum Pukht, Pavilion	₹ 104.84 Crores

9. ITC-FnB-Analytics/module2_fnb_banquet_performance.ipynb

JUPYTER NOTEBOOK

Analyzes dining plan profitability and banquet event yields. Connects directly to SQLite via SQL `JOIN` queries across `fnb_orders`, `reservations`, and `banquet_events`. Visualizes food vs beverage breakdown by signature brand and revenue contribution by event type (Weddings vs Corporate Summits).

EXECUTION STATE:	COVERS ANALYZED:	OUTPUT SIZE:
Fully Pre-Executed	121,438 covers	165 KB

3.3 Module 3: Guest Demographics & Booking Channels (`ITC-Guest-Demographics`)

10. ITC-Guest-Demographics/itc_guest_demographics_channels.csv

MODULE DATASET

Focuses on guest acquisition and loyalty dynamics. Categorizes guest nationalities into macro geographic regions (Europe, South Asia, Americas, Middle East), maps guest personas (Business/Solo, Family, Group/MICE), and segments loyalty participation across Club ITC Silver, Gold, and Platinum tiers.

DIMENSIONS:	LOYALTY TIERS:	CHANNELS:
119,390 rows × 22 cols	Club ITC Platinum/Gold/Silver	Direct, Corporate, OTA, GDS

11. ITC-Guest-Demographics/module3_guest_demographics_channels.ipynb

JUPYTER NOTEBOOK

Diagnostic analysis evaluating channel acquisition costs and cancellation risks. Demonstrates that third-party Online Travel Agencies (OTAs) carry a 41.6% cancellation rate and 15–18% commission leakage, whereas direct Club ITC loyalists display under 12% cancellations, providing a strategic foundation for direct-booking promotions.

EXECUTION STATE:	KEY INSIGHTS:	OUTPUT SIZE:
Fully Pre-Executed	Customer Acquisition Cost (CAC) vs Loyalty	145 KB

3.4 Module 4: Operating Costs & Profitability (`ITC-Profitability-Costs`)

12. ITC-Profitability-Costs/itc_operating_costs_profitability.csv

MODULE DATASET

Full financial P&L dataset conforming to the global **USALI (Uniform System of Accounts for the Lodging Industry)** framework. Computes departmental expenses: room operations, F&B cost of goods sold (COGS at 31%), distribution commissions, LEED Platinum green energy/utility costs, staff payroll, property maintenance, and marketing expenses to derive Gross Operating Profit (GOP).

DIMENSIONS:	ACCOUNTING STANDARD:	CALCULATED GOP:
119,390 rows × 21 cols	USALI 11th Edition	₹ 143.68 Crores (48.15%)

13. ITC-Profitability-Costs/module4_operating_costs_profitability.ipynb

JUPYTER NOTEBOOK

Presents executive financial analytics. Uses SQL aggregation to compute GOPPAR (Gross Operating Profit Per Available Room), depicts departmental cost distributions via horizontal bar charts, and models the profitability impact of ITC's on-site wind and solar renewable energy investments.

EXECUTION STATE:	GOPPAR REALIZED:	OUTPUT SIZE:
Fully Pre-Executed	₹ 5,612 per room night	83 KB

3.5 Master Dashboards & Database Suite (`ITC-Final-Dashboards`)

14. ITC-Final-Dashboards/itc_hotel_analytics.db

RELATIONAL DATABASE

Production-grade SQLite 3NF relational database. Contains 18 tables and views, including 87,396 cleaned transactions with all foreign keys, unique constraints, and high-performance B-tree indexes enabled. Houses both transactional tables (`reservations, invoices, payments`) and Star Schema analytical entities (`bookings, room_inventory, calendar, properties`).

FILE SIZE:	TABLE COUNT:	CLEANED RECORDS:
80.5 MB (Vacuumed)	18 Tables & Views	87,396 Bookings

15. ITC-Final-Dashboards/itc_hotels_mysql_dump.sql

MYSQL 8.0 SCRIPT

Complete standalone MySQL 8.0 DDL and sample data dump script. Formatted with `InnoDB` engine parameters, foreign key definitions, and table structures exactly matching Figures D1 and D2 of the Invertis University project report. Allows 1-click import into XAMPP, phpMyAdmin, or cloud MySQL instances.

COMPATIBILITY:	IMPORT COMMAND:	FILE SIZE:
MySQL 8.0 / MariaDB 10.4+	`mysql -u root < dump.sql`	87.6 KB

16. ITC-Final-Dashboards/data_cleaning_eda.ipynb

JUPYTER / MATPLOTLIB

The core academic data engineering notebook. Step-by-step documentation of: (1) Null value detection and imputation; (2) 31,994 duplicate removal; (3) Anomaly resolution (negative ADR and extreme outliers capped via boxplots); (4) Feature engineering; (5) 7+ Matplotlib visualizations; (6) Demonstration of live SQL aggregation queries using the Python `sqlite3` driver.

EXECUTION STATE:	CHARTS EMBEDDED:	OUTPUT SIZE:
Fully Pre-Executed with Plots	8 High-Res Figures	591 KB

17. ITC-Final-Dashboards/ITC_Executive_Live_Dashboard.xlsx

MICROSOFT EXCEL

The official Master Live Excel Dashboard. Built following the professor's strict UI/UX criteria: Features **5 bottom navigation tabs**: `Room Analytics`, `FnB Analytics`, `Demographics`, `Profitability`, and `Final`. The `Final` sheet features 4 square executive KPI cards at the top (Total Revenue, Total Profit, Avg Occupancy, F&B Sales) and embedded native Excel charts formatted with currency symbols and auto-fitted columns.

SHEET COUNT:	CHART ENGINE:	FILE SIZE:
5 Formatted Tabs	Native Openpyxl XML	19.5 KB

18. ITC-Final-Dashboards/interactive_jupyter_dashboard.ipynb

INTERACTIVE PLOTLY

Interactive web-app style executive dashboard embedded in Jupyter Notebook. Uses Plotly Dark Theme to render 4 indicator KPI cards, an interactive Donut chart for room category mix, a USALI Waterfall chart depicting revenue-to-profit conversion, and multi-bar seasonality curves.

THEME:	INTERACTIVITY:	OUTPUT SIZE:
Plotly Dark Luxury Palette	Hover, Zoom, Pan, Toggle	115 KB

19. ITC-Final-Dashboards/POWER_BI_SQL_INTEGRATION_GUIDE.md

BI ARCHITECTURE

Architectural documentation for connecting Power BI to the relational SQL database. Specifies connection options (MySQL localhost, SQLite ODBC 64-bit, and direct Python query import), the Star Schema relationship matrix (1-to-many cardinality), the complete repository of official DAX measures (ADR, RevPAR, Occupancy %, Realisation %, DSRN, DURN, MoM & WoW Growth), and blueprints for the 5 dashboard pages.

MEASURES DOCUMENTED:	DIAGRAMS:	FILE SIZE:
15+ Production DAX Formulas	Star Schema Entity Map	11.2 KB

20. ITC-Final-Dashboards/itc_luxury_theme.json

POWER BI THEME

Custom JSON theme preset for Power BI Desktop. Configures the visual canvas with ITC's signature luxury color palette: Deep Slate Background (`#0F172A`), Visual Card Fill (`#1E293B`), Imperial Gold Accent (`#D4AF37`), Emerald Profit Green (`#10B981`), and Sky Blue (`#38BDF8`).

USAGE:	FORMAT:	FILE SIZE:
View → Themes → Browse for themes	Power BI Theme JSON Specification	1.0 KB

3.6 Backend Pipeline & Code Generation Scripts (`scripts/`)

21. scripts/process_data.py

PYTHON ETL SCRIPT

The primary data engineering ETL script. Ingests raw `archive/hotel_bookings.csv`, performs currency conversion, derives realistic F&B and banquet records, applies the USALI operating expense model, and splits the data into the 4 module CSV files and the 74-column master dataset.

INPUT:	OUTPUTS:	EXECUTION TIME:
archive/hotel_bookings.csv	4 Module CSVs + 1 Master CSV	~4.8 seconds

22. scripts/create_database.py

PYTHON DATABASE BUILDER

Constructs the Third Normal Form (3NF) relational database. Performs data cleaning (missing values, 31k duplicates removed, outlier capping), creates 12 relational tables with foreign keys and unique constraints, and computes monthly aggregated KPIs in the `kpi_monthly` table using pure SQL aggregation.

DATABASE ENGINE:	OUTPUT FILE:	RECORD INGESTION:
Python SQLite3	itc_hotel_analytics.db	87,396 Transactions

23. scripts/build_pdf_schema_and_mysql.py

PYTHON SCHEMA ALIGNMENT

Aligns the database schema directly with the university report appendices (Figures D1, D2, and B1). Creates indexes on `booking_id` for sub-second query performance, populates `properties, rooms, calendar, bookings, room_inventory`, and generates the production-ready `itc_hotels_mysql_dump.sql` script for MySQL/phpMyAdmin.

EXECUTION SPEED:	OUTPUT FILES:
0.39 seconds (Optimized indexed JOINs)	itc_hotels_mysql_dump.sql, SQLite tables

24. scripts/create_excel_dashboard.py

PYTHON EXCEL BUILDER

Leverages the `openpyxl` library to programmatically generate the 5-sheet live Microsoft Excel dashboard. It writes structured KPI tables, formats cells with Indian Rupee currency styling (₹), computes formulas, and embeds native Excel Bar, Column, and Line chart XML objects into the workbook without requiring Microsoft Office.

LIBRARY:	OUTPUT FILE:	AUTOMATION:
openpyxl	ITC_Executive_Live_Dashboard.xlsx	100% Code Generated

25. scripts/generate_eda_notebook.py

PYTHON NOTEBOOK GENERATOR

Generates the standalone `data_cleaning_eda.ipynb` notebook JSON structure. Configures cells for null imputation, duplicate detection, boxplot outlier capping, and 7 Matplotlib charts, ensuring strict compatibility with Matplotlib 3.9+ (using `tick_labels` instead of deprecated parameters).

OUTPUT:	EXECUTION ENGINE:
data_cleaning_eda.ipynb	jupyter nbconvert

26. scripts/generate_notebooks.py

PYTHON NOTEBOOK GENERATOR

Generates the 4 individual module Jupyter Notebooks. Automatically injects Invertis University metadata, student team roll numbers, guide credentials, direct SQL query logic via `sqlite3`, and Seaborn/Matplotlib plotting code into each module repository.

TARGETS:	INTEGRITY:
Modules 1, 2, 3, and 4	Embedded SQL + Pandas + Matplotlib

4. Relational Database Architecture & 3NF Normalization

The database for the ITC Hotels Analytics suite conforms to **Third Normal Form (3NF)**, ensuring zero data redundancy and full referential integrity enforced via foreign keys:

TABLE / ENTITY NAME	TYPE	ROW COUNT	KEY ATTRIBUTES & RELATIONSHIPS
properties	Dimension	5	property_id (PK), property_name, category, city. Defines ITC luxury hotels.
rooms	Dimension	5	room_id (PK), room_class (Standard, Elite, Premium, Presidential, Luxury).
room_types	Dimension	11	room_type_id (PK), category_name, base_rate_inr, gst_rate_pct (12% / 18% slab).
calendar	Dimension	1,461	date_val (PK), day_name, day_of_week, month_name, month_no, quarter, year_val.
hotels	Dimension	2	hotel_id (PK), hotel_name, hotel_type (City / Resort), location, total_rooms.
staff	Dimension	10	staff_id (PK), name, role, department, hotel_name (FK), salary_inr.
users	Dimension	5	user_id (PK), username, role (Admin, Manager, Receptionist), staff_id (FK).
bookings	Fact (Analytics)	87,396	booking_id (PK), property_id (FK), check_in_date (FK), room_category, status, revenue.
room_inventory	Fact (Capacity)	9,170	(property_id, check_in_date, room_category) (Composite PK), successful_bookings, capacity.
reservations	Fact (Core)	87,396	reservation_id (PK), booking_id (UQ), hotel_name (FK), arrival_date, is_canceled, lead_time.
guests	Fact (Demographic)	87,396	guest_id (PK), booking_id (FK), country_code, region, persona, loyalty_tier.
invoices	Fact (Financial)	87,396	invoice_id (PK), booking_id (FK UQ), adr_inr, room_revenue, fnb_revenue, gross_revenue.
fnb_orders	Fact (Dining)	87,396	order_id (PK), booking_id (FK), restaurant_outlet, dining_covers, food_revenue, beverage_revenue.
banquet_events	Fact (MICE)	69,679	event_id (PK), booking_id (FK), banquet_hall, event_type, attendees, banquet_revenue_inr.
operating_costs	Fact (USALI)	87,396	cost_id (PK), booking_id (FK), rooms_cost, fnb_cogs, energy_cost, payroll, expenses, GOP.
payments	Fact (Billing)	63,371	payment_id (PK), booking_id (FK), payment_method (UPI, Card, Cash), amount_paid, status.

TABLE / ENTITY NAME	TYPE	ROW COUNT	KEY ATTRIBUTES & RELATIONSHIPS
kpi_monthly	Aggregate	26	kpi_id (PK), year, month, occupancy_rate_pct, adr_inr, revpar_inr, gross_revenue, GOP, margin.

5. Data Cleaning & Preprocessing Pipeline

To ensure strict academic rigor, the data cleaning pipeline was executed in Python and recorded in `ITC-Final-Dashboards/data_cleaning_eda.ipynb`:

CLEANING STAGE	RAW CONDITION	METHODOLOGY APPLIED	RECORDS CORRECTED
Missing Children	4 null entries in `children`	Imputed with `0` (assumed solo/couple traveler without children).	4 records
Missing Country	488 null entries in `country`	Imputed with `'Unknown'` to preserve reservation financial totals.	488 records
String 'NULL' Flags	128,933 string 'NULL' tokens in `agent` & `company`	Replaced with proper IEEE `np.nan` values for correct SQL null handling.	128,933 cells
Duplicate Transactions	31,994 identical duplicate rows	Removed via `df.drop_duplicates()`, reducing dataset to 87,396 unique stays.	31,994 rows removed
Negative ADR Error	1 transaction with ADR < 0	Data entry error replaced with column median (₹ 8,075 / €95).	1 record
Extreme Outliers	1 booking with ADR > €5,000/night	Capped at €5,000 threshold to prevent artificial skewing of variance.	1 record
Date Synthesizing	Split into year, month, day columns	Synthesized into ISO-8601 `YYYY-MM-DD` format for temporal SQL indexing.	All 87,396 records
Currency Harmonization	Historical ADR quoted in Euros	Converted to Indian National Rupee (INR) via standard baseline factor × 85.	All 87,396 records

6. The 3 Final Dashboards & Visual Architectures

6.1 Dashboard 1: Microsoft Excel Live Executive Dashboard (`.xlsx`)

- **File:** `ITC-Final-Dashboards/ITC_Executive_Live_Dashboard.xlsx`
- **Architecture:** Exactly 5 bottom tabs matching the professor's requirement: `Room Analytics`, `FnB Analytics`, `Demographics`, `Profitability`, and `Final`.
- **Executive KPI Cards:** The `Final` tab highlights 4 square cards at the top: **Total Revenue (₹ 298.42 Cr)**, **Total Profit / GOP (₹ 143.68 Cr)**, **Avg Occupancy (62.9%)**, and **F&B + Banquet Sales (₹ 104.84 Cr)**.
- **Visuals:** Native Openpyxl XML chart objects (Column charts for seasonality, Bar charts for room yield, Line charts for trend lines).

6.2 Dashboard 2: Jupyter Notebook Matplotlib & Plotly Analytics Suite

- **Files:** `ITC-Final-Dashboards/data_cleaning_eda.ipynb` & `interactive_jupyter_dashboard.ipynb`
- **Matplotlib Visualizations:** Boxplots of ADR by room category, dual-axis bar and line plots for monthly cancellations, horizontal bar charts for country distribution, and Year × Month revenue heatmaps.
- **Interactive Plotly Elements:** USALI Waterfall chart visualizing Gross Revenue → Departmental Costs → Gross Operating Profit (GOP).

6.3 Dashboard 3: Microsoft Power BI Desktop (Connected to SQL Database)

- **Files:** `POWER_BI_SQL_INTEGRATION_GUIDE.md` & `itc_luxury_theme.json`
- **Connection Engine:** Connects directly to MySQL (`itc_hotels_db`) or SQLite ODBC (`itc_hotel_analytics.db`).
- **Star Schema Model:** 1-to-many relationships connecting `dim_properties`, `dim_rooms`, and `dim_calendar` to `fact_bookings` and `fact_room_inventory`.
- **DAX Formulas Implemented:**

```
-- Occupancy Rate %
Occupancy % = DIVIDE(SUM(room_inventory[successful_bookings]), SUM(room_inventory[capacity]), 0)

-- Average Daily Rate (ADR)
ADR = DIVIDE([Total Revenue Realized], [Total Successful Bookings], 0)

-- Revenue Per Available Room (RevPAR)
RevPAR = DIVIDE([Total Revenue Realized], [Total Capacity], 0)

-- Realisation Rate %
Realisation % = DIVIDE([Total Revenue Realized], [Total Revenue Generated], 0)
```

7. System Execution & Viva Defense Guide

7.1 Quick Launch Commands on Ubuntu

```
# 1. Navigate to Project Root
cd /home/goku/ITC_Hotels_Project

# 2. Launch Interactive Terminal Hub
./run_dashboard.sh

# 3. Quick Database Inspection via SQLite CLI
sqlite3 ITC-Final-Dashboards/itc_hotel_analytics.db "SELECT count(*) FROM bookings;"

# 4. View Git Branch Structure
git branch -a
```

7.2 High-Yield Viva Q&A Defense Points

Q1: Why did you split the Kaggle dataset into 4 separate module repositories?
Answer: In enterprise hospitality environments, room revenue, culinary operations, guest CRM, and corporate accounting operate under distinct business units. Modularizing into separate repositories enforces domain separation of concerns, enables independent versioning, and complies directly with the faculty's project requirements.

Q2: Why use a 3NF Relational Database instead of querying a single flat CSV file?
Answer: A flat CSV incurs massive data redundancy, lacks foreign key integrity constraints, and slows analytical query performance. Our 3NF schema separates dimensions (Hotels, Rooms, Calendar) from transactional facts (Bookings, Invoices, Orders), eliminating update anomalies and allowing Power BI to utilize lightning-fast Star Schema 1-to-many relationships.

Q3: How is ITC's 'Responsible Luxury' reflected in the financial USALI module?
Answer: ITC Hotels is the first hotel chain globally to have all its luxury properties LEED Platinum certified. In Module 4, on-site solar and wind renewable generation caps energy/utility costs at 4.2% of revenue compared to the industry average of 8–10%, directly expanding the Gross Operating Profit (GOP) margin to a superior 48.15%.

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